

MAT133: Calculus and Linear Algebra for Commerce

Project #1: Analyzing Change in Real Data

Instruction Booklet

Contents

1 Introduction	1
2 Deadlines	1
3 Pod Project 1: Analyzing Change in Real Data	1
3.1 Finding a Data Set	1
3.2 First Draft Instructions	2
3.3 Final Report Instructions	2
3.4 Partial Grading Rubric	4
4 Appendix: Sources of Data	5

1 Introduction

Project 1 is based on the learning goals from Weeks 1 - 4. As a pod, you will be analyzing **rates of change** of a real data set of your choice, in order to address a [research question](#) of your choice.

Before getting started on this project, make sure you have read and understood the [General Project Guidelines and Instructions](#).

While all projects are holistically graded by the marking team according to their content, cohesion, accessibility, and general formatting, for Project 1 specifically, we will **pay special attention to the formatting of your Final Report**.

This document outlines specific instructions for this project, as well as a partial rubric for the focus on formatting. The Appendix contains reliable sources of data that you may wish to use for your analysis.

2 Deadlines

Each part of the project is to be submitted on Gradescope via Quercus. Note that you may re-submit as many times as you like *before the submission deadline*. If you resubmit the pod project, you'll need to re-add your pod members to your submission each time you resubmit.

The submission deadlines for the different components of the project are summarized in the table below.

Individual & Pod Contract	First Draft	Final Report & Slides	Reflection
September 29 10pm	October 6 10pm	October 13 10pm	October 20 10pm

Table 1: Summary of submission deadlines

Note that your pod's Presentation will be during your tutorial on **October 15 and 16**. Furthermore, the Reflection Piece is to be included with your regular weekly reflections, to be released on Friday, October 16.

To ensure all pod members have a chance to check that their report has been submitted properly and correct issues as needed, we will provide a 2-hour "late" submission window in Gradescope. No penalty will be applied to reports submitted or resubmitted within two hours of the deadline. However, **no project reports will be accepted past the late submission window and no project reports will be accepted via email**.

3 Pod Project 1: Analyzing Change in Real Data

In Project 1, your pod will explore a question of genuine interest using real-world data. In particular, you will use rates of change to analyze a dataset of your choice and gain new insights from the information it contains.

3.1 Finding a Data Set

Remember, this part of the project is to be completed in your pods. You will need to find a data set that you are all interested in analyzing. Please consider the following notes as you search for your data set.

- The data set should consist of at least two columns of data, with one column containing values of an input variable and at least one column containing values of an output variable.
- We recommend choosing a data set with **time** as the input variable. However, if you wish to study a data set with a different input variable, please ask your tutorial TA or ask the teaching team in a private Piazza post for advice and approval before starting. In your post, please explain why you're interested in the *rates of change* for your specific data set.
- Depending on your research question, you may wish to look at multiple data sets and combine them.
- Data sets are commonly found online in the form of a CSV file, which can be opened in Microsoft [Excel](#) (available for free to UofT students).
- A few sources are listed in the appendix, but you are not restricted to these sources. You will be required to summarize the survey or experiment that resulted in this data set.
- All pods within your hive should use their own distinct data sets! In particular, it is your job to ensure that every pod in your hive is using a different data set before you begin your analysis. You may communicate this with each other in your tutorials, Quercus group page, or through another channel of your choice.

Important: In order to genuinely answer your research question, you will need to make sure you use real data from a reliable source. While some websites contain a lot of real datasets, they may also contain fictitious datasets. You should not assume that data posted on one of the sites in the Appendix is automatically suitable! Instead, look for information on how the data was collected to verify that the data is real and reliable.

You might have questions about a data set, such as:

- Is this source reliable?
- How can I open this data in Excel?
- How can I convert this file to a format that Excel recognizes?
- What counts as a “distinct” data set, when comparing with other pods in my hive?

In this case, ask your TA during tutorials or post to Piazza in the “Projects” category. If sharing detailed information about your ideas or your report, we recommend making your post private to avoid an academic offence.

3.2 First Draft Instructions

The same instructions for the Final Report apply to the First Draft. Your pod will be given credit for submitting a complete first draft, even if it may not be fully polished. See [General Project Guidelines and Instructions](#) for more information and submission instructions.

3.3 Final Report Instructions

As you collaborate on this project, please write a report consisting of four sections that address the requirements outlined below. Aim for the final draft of your report to be both engaging and easily comprehensible to any other student in the course. Any graphs, data tables, or formulas you introduce should always be preceded by text that provides context for these items. Ensure that the report reads fluidly and appears cohesive throughout all sections. Note that you are *not* constrained to a specific number of words or pages. However, keep in mind that any extra information and/or context you include will be graded to the same standard as the rest of the report.

Caution: Do not use ChatGPT, Microsoft Copilot, or any other generative AI to generate your report. These tools can give factually inaccurate information and severely lack depth in their analyses. Writing the report with your pod (and without AI) is essential to your learning experience in this project.

1. Introduction

The purpose of this section is to introduce your project, explain the motivation for your investigation, and provide the background information needed for a reader to understand the analysis that follows.

- (a) Please begin by stating your **research question**: this is a question that you aim to answer by analyzing your data set and its rates of change. Explain why your pod is interested in answering this question. Your language here should be non-technical and understandable by the general public.
- (b) Find at least **one article or source** published by a reliable outlet (such as a government agency, academic journal, or reputable news organization) that is related to your research question. Write 2–3 sentences summarizing what this source says and how it provides context for your research question. Be sure to include a proper in-line citation and add the source to your bibliography.
- (c) **What is the meaning and relevance of your chosen data set?** Write a paragraph (or a few) to explain the meaning and relevance of your data set. Your paragraphs should address the following (in your preferred order):
 - Include a properly formatted table of values displaying the first 10 rows of your data set.
 - What do the numbers in your data set represent? Include units.
 - Which variable do you consider as the independent (or input) variable? Which is the dependent (or output) variable?
 - Where did you obtain this data set? Provide a proper in-line citation and add it to your bibliography.
 - Briefly summarize the experiment or survey that resulted in this data set.
 - Why is this data set well suited to the question your pod hopes to explore?

2. Visual Representation of Your Dataset

In this section, you will create at least one visual representation of your data set (beyond the table of values of the first 10 rows of data), using Excel or another computational tool of your choice, and describe the behaviour of the graph as it relates to your research question.

- Create a *scatterplot*, with the input variable on the horizontal axis and the output variable on the vertical axis. In this case, start by constructing a scatterplot that is interesting to your pod. You may use [Excel](#), [Octave Online](#), or any other platform of your choice.
 - Please be sure that the axes on your plot are labelled properly, including units.
 - Please **do not** include a line of best fit (or any other curve of best fit) on your scatterplot.
 - In your report, include a short paragraph **before** your graph, to introduce your graph to the reader.
 - Make sure your scatterplot and other figures are properly captioned.
- Please write a short paragraph to describe the behaviour of your function(s) based on the graph(s), as it relates to your research question. If you feel it appropriate, you may include additional visual representation(s) to accompany your paragraph (in addition to your scatterplot). Your language here should be understandable by any other student in MAT133.

3. Rates of Change and Discussion

In this section, you will analyze the rates of change for your data set and discuss how they relate to your research question. Similar to the other sections, this section should be written in a way that is easy for the reader to follow. For example, please do not include a wall of formulas without providing context. You may

organize this section as you see fit, however, please be sure to include the following elements at some point, in any order.

- Include at least one calculation of an *average rate of change* and interpret its meaning in context. (Be sure to include units in your interpretation!)
- Include at least one estimate of an *instantaneous rate of change* and interpret its meaning in context. (Be sure to include units in your interpretation!)
- Include at least one estimate of a *relative rate of change* and interpret its meaning in context. (Be sure to include units in your interpretation!)
- Create a data table consisting of (approximate) values of the instantaneous rates of change of the function represented by your data. Include a professional-looking table of the first 10 rows of this computed data in your report. Briefly describe how you calculated the numbers in the table.
- Create an approximate graph of the derivative of the function represented by your data. Construct this graph based only on your dataset and the approximations that you made using this dataset. For this graph, please do not use a formula that represents the function, even if one exists. Be sure that your axes are clearly labelled, including units.
- Provide a discussion on the rates of change of your function. Explain how they are relevant to the “real world” and how they relate to your research question. Relate your findings to the source(s) you highlighted in your introduction. Your language here should be understandable by any other student in MAT133.

Once again, please remember that your report should be engaging, comprehensible and coherent. Kindly refrain from hitting your reader in the face with a wall of formulae! Use full sentences to provide context.

4. Conclusion

In language that is easily accessible to the general public, write paragraph(s) to summarize your findings and explain how they relate to your research question.

Please ensure that your conclusion also addresses the following questions: What further questions related to this project would your pod like to investigate if you had more data? Looking at the syllabus, which upcoming topics in MAT133 might be related to these questions? Which real-world phenomenon might be related to these questions? Please identify at least one upcoming topic in MAT133 and at least one real-life phenomenon. Discuss the potential connections between these topics and your further questions.

IMPORTANT: Please make sure to pay special attention to your pod’s writing to ensure its quality and clarity. This includes integration of mathematics and figures into your paragraphs, ensuring all figures and graphs are well-labelled, and being mindful of the overall appearance of your submission. A [guide for including mathematical formulas in your reports](#) can be found on Quercus.

Please see the [General Project Guidelines and Instructions](#) for submission instructions.

3.4 Partial Grading Rubric

Overall, our marking team will assess your report based on the broad categories mentioned in the [General Project Guidelines and Instructions](#).

Additionally, for this project, the marking team will pay special attention to the **formatting** of your report according to the partial rubric outlined in Table 2 on the following page. Note that each description outlines the criteria for earning full credit in that category, but earning partial credit is also possible.

Category (points)	Description
Graphs and Tables (6)	All graphs and tables have appropriate titles, axis labels, scales, and units; all are easy to read (appropriate font, size, colour). All graphs and tables have captions and integrated appropriately into the body of the report.
Mathematical Expressions (3)	All mathematical expressions are formatted with an equation editor and integrated appropriately into the report. Calculations are woven into the narrative, with formulas explained in context rather than presented as standalone calculations.
Main Text (1)	Descriptive subtitles are included where appropriate. Text formatting (font, size, spacing) is consistent across paragraphs and sections.
Visual Consistency (1)	The same general formatting standards are applied to all parts of the project, so that the Final Report <i>looks</i> cohesive.
APA Style (1)	APA Style is accurately and consistently used throughout the report, including but not limited to: the cover page, line spacing, paragraph indentation, in-text citations, exhibit inclusions, and the bibliography.

Table 2: Partial rubric for assessing report formatting

4 Appendix: Sources of Data

The websites below contain a variety of data sets on different topics. If you have any questions regarding data sets, please reach out to us in Piazza using the “Project” category!

- [Federal Reserve Economic Data](#) - Macroeconomic data for many countries (e.g. GDP, unemployment, federal debt, and more)
- [Statistics Canada](#) - The official repository for Canadian datasets (health, population, government, etc.)
- [Yahoo Finance](#) - Price data on stocks, cryptocurrencies, currencies, and commodities
- [Data Stream](#) - Canada’s open access platform for sharing water data
- [American Economic Association data sources](#) - List of reputable sources commonly used in economics research
- [OECD](#) - Global data on various indicators related to education, environment, health, technology, and more
- [United Nations Comtrade Database](#) - Global trade statistics by products and partners
- [UCI Machine Learning Repository](#) - Data used by a wide variety of sectors
- [ICPSR](#) - Social, behavioural, and health science research data